

FAYETTEVILLE, NC, USA

WMO: 723035

Lat: 34.991N

Lon: 78.880W

Elev: 57

StdP: 100.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-3.1	-16.2	0.9	-1.0	-13.5	1.2	1.2	11.1	12.5	9.7	12.1	2.8	30	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.8	24.5	34.3	24.0	33.1	23.7	26.1	32.2	25.6	31.3	25.1	30.5	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.8	28.1	24.0	19.0	27.5	23.6	18.6	27.2	81.2	32.2	78.6	31.5	76.5	30.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.1	7.2	-9.4	38.0	2.7	1.4	-11.4	39.0	-13.0	39.8	-14.5	40.6	-16.5	41.7		
			-10.6	27.3	2.6	1.1	-12.4	28.1	-14.0	28.8	-15.4	29.4	-17.3	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	6.8	8.6	12.3	17.4	21.8	25.9	27.4	26.7	23.6	17.9	12.0	8.5
	DBStd	8.44	5.83	5.31	5.15	4.39	3.69	2.76	2.12	2.29	3.05	4.26	4.73	5.21
	HDD10.0	381	135	82	36	3	0	0	0	0	0	1	32	92
	HDD18.3	1477	359	274	196	69	13	1	0	0	3	61	195	306
	CDD10.0	3102	35	43	107	224	366	476	541	517	407	246	93	47
	CDD18.3	1157	1	2	9	41	120	226	283	259	160	47	6	3
	CDH23.3	10935	3	17	95	403	1112	2234	2931	2497	1247	352	36	9
	CDH26.7	4632	0	1	14	106	416	1031	1391	1130	458	84	3	0
Wind	WSAvg	3.2	3.4	3.5	3.8	3.9	3.4	3.1	2.9	2.6	3.0	2.9	3.0	3.1
Precipitation	PrecAvg	1189	97	89	102	76	85	111	146	138	114	76	73	78
	PrecMax	1671	199	168	225	183	254	405	370	272	374	235	162	196
	PrecMin	824	20	16	8	5	14	11	15	28	0	12	7	13
	PrecStd	180	44	44	46	41	46	69	68	64	82	53	42	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	26.0	28.4	31.6	34.8	37.0	37.3	37.3	34.2	31.5	26.7	24.6
		MCWB	17.7	18.0	17.8	20.2	22.3	24.3	25.4	24.7	22.7	22.5	19.5	18.7
	2%	DB	21.1	22.6	26.1	29.3	32.6	34.9	35.7	35.2	32.5	28.9	24.0	22.1
		MCWB	16.1	15.7	17.1	19.0	21.6	23.8	25.0	24.5	23.0	21.2	17.7	17.8
	5%	DB	18.7	20.2	23.8	27.7	31.0	33.6	34.1	33.6	31.2	27.1	22.4	19.8
		MCWB	14.9	14.4	15.6	18.1	21.2	23.4	24.6	24.3	22.8	19.9	17.1	16.7
	10%	DB	16.4	17.8	21.3	25.9	28.9	32.2	32.8	32.2	29.4	25.1	20.3	17.4
		MCWB	13.0	12.9	14.5	17.4	20.5	23.0	24.3	24.1	22.2	19.6	16.1	14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.5	20.2	22.4	24.1	26.0	27.3	26.8	25.6	24.7	21.6	20.6
		MCDB	21.5	23.6	24.6	27.1	30.2	32.5	33.7	32.7	30.6	26.9	23.4	22.9
	2%	WB	17.5	17.8	18.8	21.0	23.2	25.3	26.3	25.9	24.7	23.2	19.8	18.9
		MCDB	19.7	20.9	23.1	26.1	29.1	31.6	32.6	31.7	29.3	26.6	22.3	21.0
	5%	WB	15.6	15.8	17.6	20.0	22.4	24.7	25.7	25.4	24.1	22.0	18.5	17.0
		MCDB	18.0	19.1	21.1	25.1	28.1	30.6	31.6	31.0	28.3	25.1	21.2	19.2
	10%	WB	13.4	13.7	16.3	18.9	21.7	24.1	25.2	24.9	23.5	20.7	17.0	14.8
		MCDB	15.9	17.1	20.5	23.9	26.9	29.6	30.8	30.0	27.1	24.0	19.6	17.2
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.6	12.3	12.6	11.8	10.9	10.4	10.3	10.1	11.5	12.0	10.7
		MCDBR	13.1	14.3	14.7	14.5	13.5	12.5	11.7	12.0	12.0	12.7	13.5	12.4
		MCWBR	8.9	9.0	7.4	6.2	4.7	3.8	3.7	3.5	4.2	5.8	8.0	8.5
	5% WB	MCDBR	11.5	13.0	12.3	11.9	11.2	10.9	10.4	10.4	9.5	9.4	10.8	11.1
		MCWBR	9.2	9.7	7.9	6.2	4.6	3.9	3.8	3.6	3.9	5.1	7.9	8.9
Clear-Sky Solar Irradiance	taub	0.322	0.334	0.370	0.408	0.466	0.501	0.548	0.524	0.432	0.392	0.343	0.319	
	taud	2.481	2.439	2.373	2.264	2.146	2.090	1.979	2.035	2.290	2.367	2.463	2.523	
	Ebn at Noon	882	909	901	878	828	796	756	766	827	832	848	864	
	Edh at Noon	85	99	115	134	153	162	180	167	124	105	86	77	
All-Sky Solar Radiation	RadAvg	2.66	3.33	4.40	5.52	6.05	6.37	6.06	5.46	4.62	3.80	2.93	2.33	
	RadStd	0.16	0.36	0.33	0.37	0.47	0.42	0.38	0.28	0.38	0.47	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+126		
(47)	Regional (65 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+113		

Nomenclature: See separate page